

MyDesign Portfolio

Element H

Students: The template below is one way to structure your testing plan, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own testing plan, be sure to delete the instructions and delete the examples given in the template table.

Guidance before filling out the table on the next page:

- Enter *all* of the high-priority design requirements you listed in Element C into the table below.
- Determine an appropriate testing method for each requirement. For example, the test may involve measurement, inspection, physical actions, or mathematical modeling.
- As you plan, use your scientific method best practices, e.g. creating experimental controls, defining independent and dependent variables, only changing one variable at a time and changing it in increments to find a trend, state your prediction, etc.
- Your testing plan for a given requirement must include enough detail so that a person who is not on the design team could fully replicate the test. Include photos and/or diagrams to add clarity. Each test should include the following details:
 - how the test will be set up
 - what needs to happen during the test
 - test cases (if applicable)
 - number of trials of the test expected (aim for 7 or more trials)
 - the expected range of results that could be recorded, including units
 - the criteria for passing or failing the associated design requirement, including any permissible range for error and/or minimal thresholds
 - any goals your team has set beyond the passing result

Element H: Prototype Testing and Data Collection Plan

Testing Details (H1, H2, H3, H4)

	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1	Our design solution includes a covering for the cables to be within the 18in width and 236in length by 3 in height of the walkway to prevent spacing problems.	1) Insert our cable trough design into the walkway full of wires 2) we then measured to see if our design fit within the measurements	Test 1: successful as it had fit in our dimensions	Parts of the prototype may: - be too large - be too small	Fail: one or more parts of the prototype is too large/small Pass: zero parts of the prototype are too large/small Goal: all parts of the prototype fit perfectly
2	Our design requires the cover for the cables to be strong enough for the average sized male or female to step, stand, and	1) Put the cable trough on the floor and have people stand and walk on it 2) Have a small-sized build person stand and walk across it 3) Then the same for a middle sized build 4) Finally someone with a bigger sized build	Test 1: Successful, The Cable Trough withstood the weight of the smallest build person. Test 2: successful, CT held the middle sized person Test 3: successful, CT held the largest	Parts of the prototype may: - Start sagging or crack under stress - Break or loosen the glue	Fail: parts may break Pass: zero parts break Goal: all parts stay intact and work well

	potentially walk on them		sized person		
3	Our design needs to have cable covers to be able to snap together into one large assembly and function but also be used in individual and smaller units.	1) Put the cable connectors together and test their grip 2) Shake the CTs a little and see if they still hold	Test 1: successful the CT connectors held through mild and harsh vibrations and shaking	Parts of the Prototype may: - Fall off - Detach and warp	Fail: the CT connectors fall off Pass: the CT connectors stay on Goal: the CT connectors stay on and remain their grip on the CTs
4	Our design must require enough stability so that the average person that walks upon the design, doesn't harm themselves and/or the design which protects the cables. Miniature peg holders will be used to keep the design firm and still.	1) Have a person walk on it 2) Have them stand and move around on it to test it's stability	Test 1: successful, The CT didn't wobble much as people vigorously moved on it and it remain still it'sfrom it's grip	Parts of the prototype may: - Get loose - Snap off	Fail: pegs get loose and fall off Pass: all parts stay secure and functional Goals: The CT remains intact fully and stay s secure.

5	Our design must be efficient enough to be easily carried or be put away in storage if it is no longer needed.	1) Stack the CTs in various different ways and test their wobbly-ness when walking with them.	Test one: successful, the CT where stack and walked around with them at various speed, and remained stable.	Prototypes may: - all over	Fail: CTs fall over when in storage and when someone's holding them Pass they stay stable in all situations Goals: for the CTs to remain stable and not fall over anytime.
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Feedback From Field Experts (H5)

Guidance to students about this section of Element H.

- a) "Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.
- b) You must provide evidence in this section that your testing plan has been reviewed by 2-3 field experts.
- c) Do not incorporate their feedback in the table above. Rather, make a copy of the table in the section above, paste it into the section below, and make all revisions only in the table below. This provides evidence that you incorporated the feedback and shows your testing plan's progression.

Revised Testing Details That Incorporate Field Expert Feedback (H5)

Walpole Engineers Design-a-Thon - Judge's Feedback			
Judge: <u>Breccini</u>		Team/Project Title: <u>Walkway</u>	
Proficient (3)	Developing (2)	Novice (1)	Score
Learning Goal: Identify and describe a problem that can be solved with a potentially new product or process. - Can you explain what the problem is and the process you used to come to that conclusion? - What kind of input/feedback did your primary stakeholder provide for the problem and/or design requirements? - How did empathy play a role in determining the problem and design requirements? - How did you decide which design requirement was the most important to focus on?			2
Learning Goal: Plan and conduct research by gathering relevant and credible data, facts, and information. - What evidence is there that your current design is based on your observations and interview of your primary stakeholder? - Where did you draw inspiration from for your current design? - What has your primary stakeholder already tried? How did you apply this information to your current design?			3
Learning Goal: Articulate appropriate STEM practices and principles in the design. What kind of research did you do to learn more about the problem and how did you apply that to your current design?			2
Learning Goal: Evaluate solution alternatives and select a final design by considering assumptions, trade offs, criteria, and constraints. - What other ideas did you come up with during the brainstorming phase? - How did you make the decision to focus on your current design? - Can you walk through all the different phases/versions that your current design/prototype went through? - What did you learn or take away from all the previous versions or attempts? - Do you have any pictures or physical models of your previous or early attempts? - What do you really like about your current design/prototype? - What are you looking forward to improving in the next version?			3
Learning Goal: Create a prototype. - How did you make your current prototype? - What challenges did you face, and how did you overcome them? - What do you really like about your current design/prototype? - What are you looking forward to improving in the next version? - Were there any limitations or constraints that impacted what you were able to build?			3
Learning Goal: Create and implement a Test Plan to evaluate the performance of the design solution. - What have you learned or what are you hoping to learn from testing your prototype today? - Who is your intended user and how did you consider their input in your Test Plan? - Can you measure any quantifiable constraints from your Design Brief? If so, what is an acceptable measure or result?			3
Learning Goal: Use various engineering communication methods.			2
Judge's Comments: Good use of Plastic Maze Concept was clear!			18

Walpole Engineers Design-a-Thon - Judge's Feedback

Judge: Jim Love Team/Project Title: Walkway Cable

Proficient (3)	Developing (2)	Novice (1)	Score
Learning Goal: Identify and describe a problem that can be solved with a potentially new product or process. - Can you explain what the problem is and the process you used to come to that conclusion? - What kind of input/feedback did your primary stakeholder provide for the problem and/or design requirements? - How did empathy play a role in determining the problem and design requirements? - How did you decide which design requirement was the most important to focus on?			
<u>met w/ Mr D & saw the problem firsthand</u>			3
Learning Goal: Plan and conduct research by gathering relevant and credible data, facts, and information. - What evidence is there that your current design is based on your observations and interview of your primary stakeholder? - Where did you draw inspiration from for your current design? - What has your primary stakeholder already tried? How did you apply this information to your current design?			
<u>new attempt so no previous designs to draw from</u>			3
Learning Goal: Articulate appropriate STEM practices and principles in the design. What kind of research did you do to learn more about the problem and how did you apply that to your current design?			
<u>in studio</u>			3
Learning Goal: Evaluate solution alternatives and select a final design by considering assumptions, trade offs, criteria, and constraints. - What other ideas did you come up with during the brainstorming phase? - How did you make the decision to focus on your current design? - Can you walk through all the different phases/versions that your current design/prototype went through? - What did you learn or take away from all the previous versions or attempts? - Do you have any pictures or physical models of your previous or early attempts? - What do you really like about your current design/prototype? - What are you looking forward to improving in the next version?			
<u>I didn't see your sketches?</u>			2
Learning Goal: Create a prototype. - How did you make your current prototype? - What challenges did you face, and how did you overcome them? - What do you really like about your current design/prototype? - What are you looking forward to improving in the next version? - Were there any limitations or constraints that impacted what you were able to build?			
<u>acrylic w/ 3D printed legs</u>			3
Learning Goal: Create and implement a Test Plan to evaluate the performance of the design solution. - What have you learned or what are you hoping to learn from testing your prototype today? - Who is your intended user and how did you consider their input in your Test Plan? - Can you measure any quantifiable constraints from your Design Brief? Also, what is an acceptable measure or result?			
<u>testing ongoing to find proper placement legs</u>			3
Learning Goal: Use various engineering communication methods.			
<u>good posters, CAD, slides</u>			3
Judge's Comments: <u>Love the minimalist design! see-through</u>			